

PAPER_777: NGC 6217 — UQFF Barred Spiral Galaxy Standard Rotation

Author: Daniel T. Murphy

Framework: UQFF (Universal Quantum Field Superconductive Framework)

Session: 181 | v5.41

Date: 2026

CP4 Class: #361 — NGC6217BarredSpiralUQFFCalculator

Abstract

NGC 6217 is a barred spiral galaxy ~67 million light-years distant ($z = 0.0045$) in the constellation Ursa Minor. It became notable as one of the first targets imaged by the Hubble Space Telescope's repaired Wide Field Camera 3 (WFC3) after the 2009 Servicing Mission 4, demonstrating the camera's restored capability. With moderate star formation ($\text{SFR} \approx 1 \text{ M}_\odot/\text{yr}$) and an extended rotating disk of $\sim 10^{11} \text{ M}_\odot$, NGC 6217 represents a typical SBbc barred spiral. Under UQFF, standard rotation velocity ($v = 10^5 \text{ m/s}$) and typical HII B-field yield $g_{\text{NGC6217}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$.

1. Introduction

NGC 6217 (also catalogued as UGC 10470) has an apparent magnitude of ~ 11.2 , a disk diameter of $\sim 30,000 \text{ ly}$, and a Hubble morphological type of SBbc (barred spiral, late intermediate). Its bar structure funnels gas toward the central region, sustaining moderate star formation. The galaxy's selection as a Hubble Servicing Mission 4 first-light target in 2009 reflects its moderate brightness and interesting bar+ring morphology at $z = 0.0045$ ($\sim 67 \text{ Mly}$). UQFF treats it as a standard barred galaxy with SFR coupling through the galactic bar channel.

2. Master UQFF Gravity Equation

$$g_{\text{NGC6217}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 + M_{\text{sf}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{11} \text{ M}_\odot = 1.989 \times 10^{41} \text{ kg}$	Estimate
Disk radius	r	$3 \times 10^{20} \text{ m}$ ($\sim 30 \text{ kly}$)	NED
SFR	—	$1 \text{ M}_\odot/\text{yr}$	Moderate SBbc
Age (evolution time)	t	$5 \times 10^9 \text{ yr} = 1.578 \times 10^{17} \text{ s}$	Hubble time
M_{sf}	—	0.045	UQFF SFR integral

Parameter	Symbol	Value	Source
Redshift	z	0.0045	NED spectroscopic
v_{EM}	v	10^5 m/s	Disk rotation
B_{EM}	B	10^{-5} T	Galactic field
f_{TRZ}	—	0.04	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$\begin{aligned}
 g_{grav} &= G \times M / r^2 \\
 &= 6.6743e-11 \times 1.989e41 / (3e20)^2 \\
 &= 1.328e31 / 9e40 \\
 &= 1.476e-10 \text{ m/s}^2
 \end{aligned}$$

Step 2: Cosmic Expansion Factor

$$\begin{aligned}
 H(z) &= H_0 \times (1 + z)^{(3/2)} \approx 2.315e-18 \text{ s}^{-1} \\
 H(z) \times t &= 2.315e-18 \times 1.578e17 = 0.3653 \\
 1 + H(z) \times t &= 1.3653
 \end{aligned}$$

Step 3: Star-Formation Mass Fraction (Bar-Channeled)

$$\begin{aligned}
 \text{Bar-driven SFR} &= 1 \text{ M}_\odot/\text{yr, integrated over 5 Gyr} \\
 M_{sf} &= 0.045 \text{ (UQFF bounded, } \sim 4.5\% \text{ mass fraction in young stars)} \\
 1 + M_{sf} &= 1.045
 \end{aligned}$$

Step 4: Time-Reversal Correction

$$\begin{aligned}
 f_{TRZ} &= 0.04 \text{ (moderate spiral, bar-driven)} \\
 1 + f_{TRZ} &= 1.04
 \end{aligned}$$

Step 5: Gravitational Total

$$\begin{aligned}
 g_{grav_total} &= 1.476e-10 \times 1.3653 \times 1.045 \times 1.04 \\
 &= 1.476e-10 \times 1.488 = 2.197e-10 \text{ m/s}^2
 \end{aligned}$$

Step 6: Aether Electromagnetic Correction

$$\begin{aligned}
 v &= 10^5 \text{ m/s (disk rotation velocity)} \\
 B &= 10^{-5} \text{ T (galactic B-field)} \\
 a_{EM} &= (e/m_p) \times (v \times B) \times \Lambda_{UQFF} \\
 &= 9.575e7 \times (10^5 \times 10^{-5}) \times 11 \times 10^{-12} \\
 &= 9.575e7 \times 1 \times 1.1e-11 \\
 &= 1.053e-3 \text{ m/s}^2
 \end{aligned}$$

Step 7: Final Solution

$$\begin{aligned} g_{\text{NGC6217}} &= 2.197\text{e-}10 + 1.053\text{e-}3 \\ &\approx 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

The classical gravitational term ($2.197 \times 10^{-10} \text{ m/s}^2$) is 7 orders of magnitude smaller than the Aether EM result ($1.053 \times 10^{-3} \text{ m/s}^2$). The UQFF result thus captures disk rotation dynamics through the electromagnetic Aether coupling, not Newtonian gravity. The bar structure in NGC 6217 channels gas inward, sustaining the moderate SFR = $1 \text{ M}_{\odot}/\text{yr}$ and bar-enhanced $M_{\text{sf}} = 0.045$, slightly higher than a purely symmetric flocculent spiral of similar mass. As with NGC 2841, the dominant physics is electromagnetic at these rotation velocities.

5. UQFF Framework Advancement

- NGC 6217 validates standard SBbc barred spiral UQFF field result
 - Bar-driven SFR = $1 \text{ M}_{\odot}/\text{yr}$ produces $M_{\text{sf}} = 0.045$ (canonical bar value)
 - Hubble SM4 first-light observation history preserved in UQFF canon
 - SBbc result consistent with other moderate barred spirals (NGC 2841, NGC 1300)
-

6. Conclusions

UQFF applied to NGC 6217 yields $g_{\text{bar_spiral}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$, confirming standard barred galaxy behavior. The Hubble-celebrated target joins the UQFF canon as the SBbc benchmark alongside Milky Way analogs. Bar-enhanced gas flow, expressed through $M_{\text{sf}} = 0.045$ and moderate SFR, demonstrates how UQFF differentiates galaxy morphological types within a unified framework.

PAPER_777, CP4 class #361. v5.41.